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New Late Jurassic palaeomagnetic data from the northern Sichuan basin: implications for the deformation of the Yangtze craton

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SUMMARY

Upper Jurassic red sandstones and red siltstones were collected from 67 layers at 12 localities in the Penglaizhen formation. This formation is in the north of Bazhong county (31.8°N, 106.7°E) in the Sichuan basin, which is located in the northern part of the Yangtze craton. Thermal demagnetization isolated a high-temperature magnetic component with a maximum unblocking temperature of about 690 °C from 45 layers. The primary nature of the magnetization acquisition is ascertained through the presence of magnetostratigraphic sequences with normal and reversed polarities, as well as positive fold and reversal tests at the 95 per cent confidence level. The tilt-corrected mean direction of 36 layers is $D = 20.0°$, $I = 28.8°$ with $\alpha_{95} = 5.8°$. A Late Jurassic palaeomagentic pole at 64.7°N, 236.0°E with $A_{95} = 7.0°$ is calculated from the palaeomagnetic directions of 11 localities. This pole position agrees with the two other Late Jurassic poles from the northern part of the Yangtze craton. A characteristic Late Jurassic pole is calculated from the three poles (68.6°N, 236.0°E with $A_{95} = 8.0°$) for the northern part of the Yangtze craton. This pole position is significantly different from that for the southern part of the Yangtze craton. This suggests that the southern part of the Yangtze craton was subjected to southward extrusion by 1700 ± 1000 km with respect to the northern part. Intracraton deformation occurred within the Yangtze craton.

Key words: continental deformation, Jurassic, palaeomagnetism, tectonics.

1 INTRODUCTION

The Asian continent is a mosaic of tectonic blocks. The eastern half of China is composed of two major blocks: the North China Block (NCB) and the South China Block (SCB) (Fig. 1a). The SCB comprises the Yangtze craton, the Huanan fold belt and the Maritime fold belt (Zhang *et al.* 1984).

There is controversy over the time of amalgamation of the NCB and the Yangtze craton. Preliminary palaeomagnetic studies claimed that the two blocks sutured until Middle Jurassic times (Lin *et al.* 1985, Yang *et al.* 1992), whereas geological investigations in the Qinling-Dabie Shan region suggest that the collision of the two blocks occurred in Devonian times (Mattauer *et al.* 1985) and that tectonic deformation within the Qinling–Dabie Shan region continued until the Early Cretaceous (e.g. Ziegler *et al.* 1996).

Recently, reliable palaeomagnetic data with a positive fold test have been obtained from the Late Jurassic rocks of both the NCB and the Yangtze craton. Gilder & Courtillot (1997) reported on Late Jurassic data from Anhui province in the southern part of the NCB, while Huang & Opdyke (1991) obtained a reliable Late Jurassic pole from the Huili area in the southern part of the Yangtze craton. Although these workers concluded that the suturing of the two blocks was completed by the Middle to Late Jurassic boundary, these Late Jurassic poles from the two blocks occupy significantly different positions (Fig. 1c). This discrepancy could result from a later suturing between these two blocks: did the suturing continue until the Early Cretaceous? Alternatively, tectonic deformation within the Yangtze craton cannot be completely excluded.

Reliable palaeomagnetic data from the northern part of the Yangtze craton, located between Anhui province and Huili, are required in order to understand the origin of the discrepancy between these two poles. Tectonic intercraton and intracraton deformation will be discussed by comparing our Late Jurassic pole for the northern part of the Yangtze craton with the poles for Huili in the southern part of the Yangtze craton and for Anhui province of the NCB.

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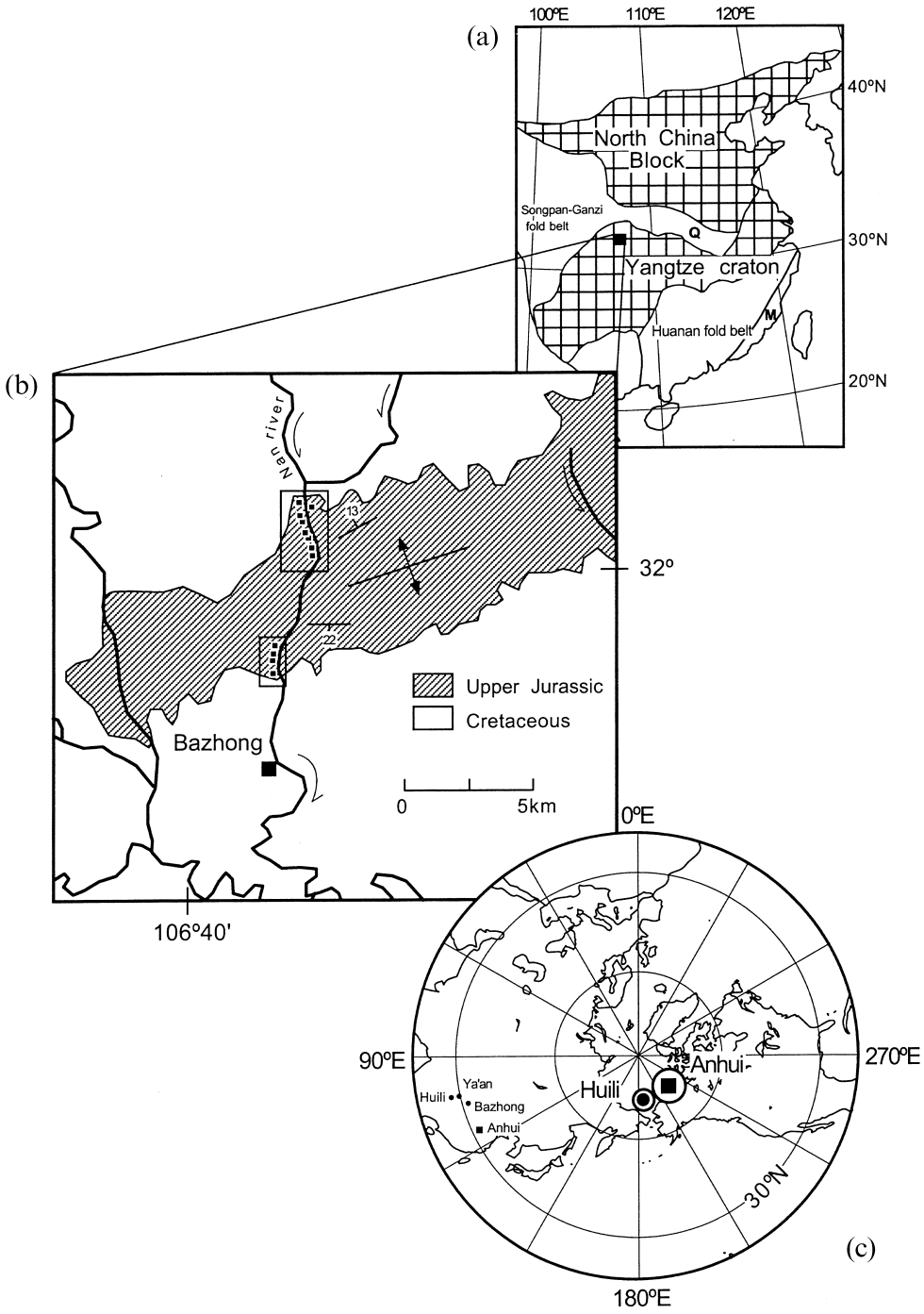


Figure 1. (a) Simplified tectonic map of the eastern half of China (based on Zhang *et al.* 1984). Q = Qinling accretionary fold belt; M = Maritime fold belt. (b) Geological sketch map of the sampling areas in the Yangtze craton. The two boxes on the Upper Jurassic formation are sampling areas on the northern and southern limbs of an anticline. Solid squares indicate sampling localities. Typical attitudes of strata are shown. (c) Late Jurassic palaeomagnetic poles for Anhui (a solid square) and Huili (a solid circle), with their circles of 95 per cent confidence.

2 GEOLOGICAL SETTING AND SAMPLING

The Sichuan basin is a large sedimentary basin located in the northern part of the Yangtze craton. This basin is mostly filled with Mesozoic continental redbeds. Upper Jurassic and Lower Cretaceous redbeds in this basin are gently folded. The folding took place after Palaeogene times, because the early Neogene formation uncomformably overlies Jurassic to Palaeogene strata in the northwestern margin of the Sichuan basin (Bureau of Geology and Mineral Resources of Sichuan Province 1997).

According to the Bureau of Geology and Mineral Resources of Sichuan Province (1991), Upper Jurassic strata in the Sichuan basin are subdivided into the Suining and the Penglaizhen formation's in ascending order. The Penglaizhen formation consists of sandstones and mudstones. It disconformably underlies the Lower Cretaceous Chengqiannyan Group. The Penglaizhen formation is assigned to the Late Jurassic on the basis of

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abundant fossils of ostracods such as *Darwinula oblonga* and *Darwinula leguminella* (Bureau of Geology and Mineral Resources of Sichuan Province 1982).

The Penglaizhen formation forms an anticline with a northeast-trending axis in the north of Bazhong county (31.8°N, 106.7°E) (Fig. 1b). Our palaeomagnetic study focused on red sandstones and red siltstones on both the northern and the southern limbs of the anticline. Samples were collected from 44 layers at eight localities on the northern limb, and from 23 layers at four localities on the southern limb. One to seven hand samples were collected from each layer and oriented with a magnetic compass. Sampling at each locality covered up to 10 m of stratigraphic thickness contained within four to seven layers. The present declination value at the sampling locality was estimated from calculations using the International Geomagnetic Reference Field, 1995 (IAGA Division V, Working Group 8, 1995).

3 PALAEOMAGNETIC RESULTS

One to five core samples were prepared from each oriented sample, and cut into specimens 25 mm in diameter and 23 mm in height. The remanent magnetization of a specimen was measured with a 2G cryogenic magnetometer. Stepwise thermal demagnetization up to 690 °C was conducted with a Natsuhara TDS-1 thermal demagnetizer for all specimens. The acquisition of isothermal remanent magnetization (IRM) and back-field demagnetization were performed on a 2G pulse magnetizer up to maximum fields of 2.7 T. Thermal demagnetization of the composite IRMs (Lowrie 1990) was conducted with a 2G pulse magnetizer and the Natsuhara TDS-1 demagnetizer. Magnetic components were separated using principal component analysis (Kirschvink 1980).

The magnetic properties seem to depend on lithology. The mean NRM (natural remanent magnetization) intensity becomes weaker as the grain size in specimens increases [8.6 ± 7.8 (10^{-3} A m^{-1}) for siltstones, 3.8 ± 0.28 (10^{-3} A m^{-1}) for silt-sandstones, 2.6 ± 1.6 (10^{-3} A m^{-1}) for fine-grained sandstones, and 2.3 ± 7.8 (10^{-3} A m^{-1}) for sandstones]. During thermal demagnetization, specimens with finer grains (siltstones, silt-sandstones and fine-grained sandstones) generally revealed a stable high-temperature component. This component appeared above 350 °C in siltstones (Figs 2a and b) and above 650 °C in silt-sandstones and fine-grained sandstones (Figs 2c and d). Specimens of sandstones revealed no stable high-temperature magnetization component (Figs 2e and f). After the demagnetization step of 350 °C, the endpoints of the magnetization vector were randomly scattered.

The high-temperature component has a maximum unblocking temperature of about 690 °C, indicating that this component is carried by haematite. This indication is confirmed by IRM and thermal demagnetization of the composite IRMs experiments (Fig. 3): the IRM did not reach saturation by 2.7 T and showed a coercivity of remanence of about 0.4 T during back-field demagnetization. The magnetization of all three components of the composite IRMs remained until 680 °C. We recognize the high-temperature component of haematite as a characteristic remanent magnetization (ChRM).

A palaeomagnetic direction for each layer is calculated from the ChRM directions of specimens in the layer (Table 1). 39 layers have more than three specimens with a ChRM: the circle of 95 per cent confidence is less than 40° except for three layers. Six layers have one or two specimens with a ChRM. No palaeomagnetic directions are defined for the sandstone layer, because sandstone specimens have no ChRM. The palaeomagnetic directions are obtained from 31 of 44 layers on the northern limb and from 14 of 23 layers on the southern limb (Fig. 4).

Each layer contains specimens with only one polarity: normal polarities are obtained in 28 layers and reversed polarities in 17 layers (Fig. 4). We define a layer with a north-seeking (south-seeking) palaeomagnetic direction as a layer with a normal (reversed) polarity. A magnetostratigraphic sequence with both reversed and normal polarities is discovered at three localities (BZ07, BZ11 and BZ17). Six magnetostratigraphic polarity zones are observed within about 1100 m of stratigraphic thickness on the northern limb, and four zones within about 600 m of stratigraphic thickness on the southern limb.

A mean direction for the Penglaizhen formation is calculated by assigning unit weight to the palaeomagnetic direction of a layer. Two mean directions based on data sets of all layers and selected layers are listed in Table 1. A mean direction is also calculated by assigning unit weight to the palaeomagnetic direction of a sampling locality. The same values are obtained from the three directions within the 95 per cent confidence level (Table 1).

The mean direction for the selected layers is calculated from 36 out of 45 layers. Data sets of nine layers are discarded, because the number of specimens in six layers is too small (less than two) to evaluate the dispersion of the palaeomagnetic direction of a layer, and three layers have a circle of 95 per cent confidence larger than 88.0°. The mean direction is $D = 22.4°$, $I = 28.6°$ with $\alpha_{95} = 7.2°$, $k = 11.8$ *in situ*, whereas it is $D = 20.0°$, $I = 28.0°$ with $\alpha_{95} = 5.8°$, $k = 18.1$ after tilt correction (Fig. 5). We recognize this mean direction for the selected layers as a characteristic palaeomagnetic direction of the Penglaizhen formation.

4 DISCUSSION

We apply the fold test of McElhinny (1964) and McFadden (1990) to examine a pre-folding origin for the characteristic palaeomagnetic direction of the Penglaizhen formation. On applying progressive unfolding with a step of 10 per cent, the precision parameter (k) reaches a maximum at 100 per cent unfolding. After 100 per cent tilt correction, the precision parameter (k) increases from 11.8 to 18.1, indicating that the fold test of McElhinny (1964) is positive at the 95 per cent confidence level. Based on McFadden's formula (McFadden 1990), the calculated value (ξ_1) is 7.52 *in situ* and 3.20 after tilt correction, while the critical value (ξ) is 6.98 at the 95 per cent confidence level. This indicates that the fold test is positive at the 95 per cent level. We conclude that the tilt-corrected mean direction is of pre-folding origin.

Because the Penglaizhen formation forms a magnetostratigraphic sequence with normal and reversed polarities, the characteristic remanent magnetization was probably acquired through a depositional process. The palaeomagnetic direction of the Penglaizhen formation is considered to be of primary origin.

The tilt-corrected mean directions of normal and reversed polarities pass the reversal test at the 95 per cent confidence

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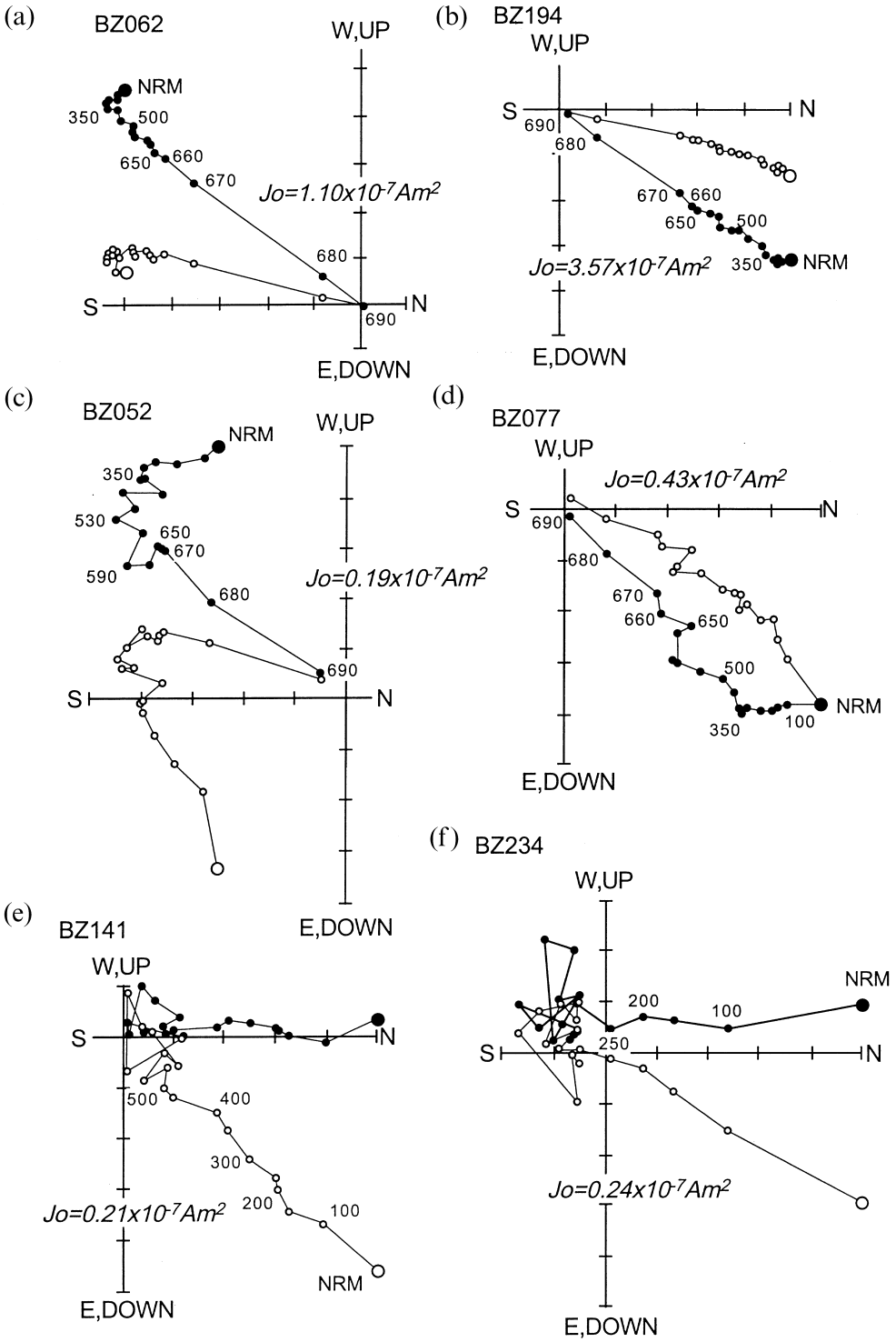


Figure 2. Thermal demagnetization diagrams (*in situ* coordinates). Temperature steps are in °C. Orthogonal projections of demagnetization are shown for (a, b) siltstones (layers BZ05-*2* and BZ19-*4*), (c) silt-sandstones (BZ05-*5*), (d) fine-grained sandstones (BZ07-*4*) and (e, f) sandstones (BZ13-*1* and BZ23-*5*).

level (McFadden & McElhinny 1990): the angular distance is 8.7° (Table 1), whereas the critical angles is 11.9°. The tilt-corrected mean direction of the 36 layers therefore represents the palaeomagnetic direction for a fairly long period in the Late Jurassic in the Bazhong area.

The Late Jurassic pole for the Penglaizhen formation of Bazhong is compared with poles for the northern part of the Yangtze craton (Fig. 6a). The Bazhong pole is calculated from the palaeomagnetic directions of 11 localities (Table 1), because the previously reported poles were based on the palaeomagnetic directions of sites (Lin *et al.* 1985; Enkin *et al.* 1991; Bai *et al.* 1998). The Bazhong pole lies at 64.7°N, 236.0°E with $A_{95} = 7.0°$. Enkin *et al.* (1991) and Bai *et al.* (1998) obtained reliable Late Jurassic poles from Ya'an (29.9°N, 102.9°E) and

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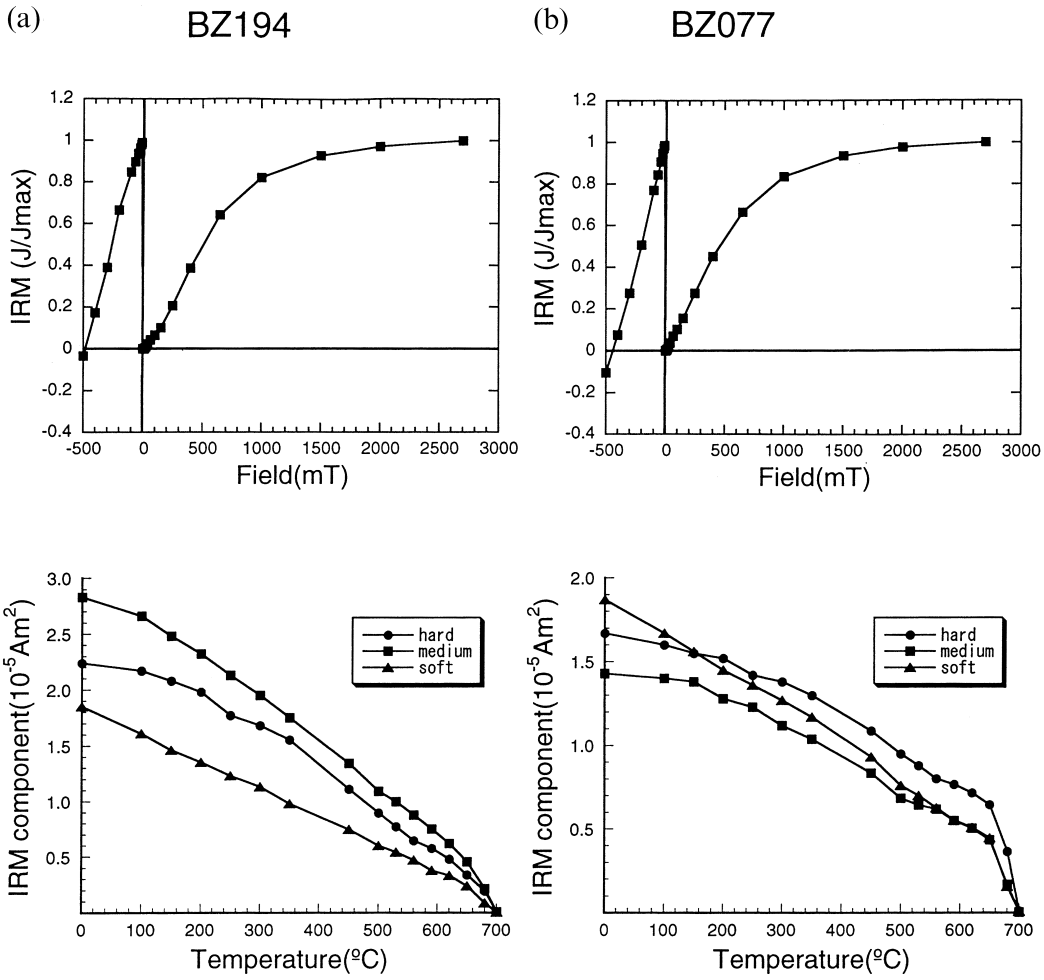


Figure 3. The normalized isothermal remanent magnetization (IRM) acquisition curve, its back-field demagnetization curve, and the thermal demagnetization curve of the composite IRM. Each component of the IRM was produced by applying a different DC field (2.7, 0.4 and 0.12 T) to three perpendicular axes of a specimen. (a) siltstones (BZ19-*4*) and (b) fine-grained sandstones (BZ07-*4*).

Wangchang (32.1°N, 106.1°E) (see Table 2): both palaeomagnetic data sets passed the fold test. The pole for Ya'an is corrected after removing the local rotation as postulated by Enkin *et al.* (1991). Although one pole is reported from Zhejiang (29.4°N, 120.0°E) by Lin *et al.* (1985), we rejected it because no field test was applied to the palaeomagnetic directions. The Bazhong pole is indistinguishable from the poles for Ya'an and Wangchang at the 95 per cent confidence level. This indicates that no relative motion occurred within the northern part of the Yangtze craton. Based on the three poles for Bazhong, Ya'an and Wangchang, we obtain a characteristic Late Jurassic pole position for the northern part of the Yangtze craton (68.6°N, 236.0°E with $A_{95} = 8.0°$).

The pole for the northern part of the Yangtze craton is compared with reliable poles for the southern part of the Yangtze craton (Fig. 6b). Only one reliable Late Jurassic pole has been obtained for Huili (26.8°N, 102.3°E), by Huang & Opdyke (1991, 1993). These authors proposed the pole based on palaeomagnetic directions from 20 sites which passed both the fold and the reversal tests. Although Zhu *et al.* (1988) reported a pole for Dukou (26.2°N, 101.5°E), their pole is not accepted as reliable because the palaeomagnetic data set was obtained without sufficient demagnetization and without the fold test. The pole for the northern part of the Yangtze craton is significantly different from the pole for Huili. We have discovered a significant discrepancy between the pole positions for the northern and the southern parts of the Yangtze craton.

This discrepancy is ascribed to tectonic deformation within the Yangtze craton. Since the two poles for the northern and the southern parts of the Yangtze craton are calculated on the basis of the palaeomagnetic directions, which cover a fairly long period, this discrepancy can hardly be explained by palaeomagnetic behaviour (e.g. excursion). The southern part of the Yangtze craton experienced southward displacement with respect to the northern part of the Yangtze craton. The observed inclination at Huili corresponds to a palaeolatitude of $27.0 \pm 3.9°$N, whereas the expected palaeolatitude at Huili is $11.2° \pm 8.0°$N (the expected value is calculated from the characteristic Late Jurassic pole for the northern part of the Yangtze craton). Southward displacement by $15.8° \pm 8.9°$ is expected for the Huili area, which corresponds to 1700 ± 1000 km with respect to the northern part of the Yangtze craton. No significant rotational motion is detected: $1.9 \pm 13.9°$.

The southern part of the Yangtze craton on which the Huili area is situated seems to have behaved independently of the main part of the Yangtze craton. The southern part of

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Table 1. Palaeomagnetic results for redbeds in the Upper Jurassic Penglaizhen formation.

Locality	Lat. (°N)	Long. (°E)	layer	n/N	In-situ Dec. (°)	In-situ Inc. (°)	Tilt-corrected Dec. (°)	Tilt-corrected Inc. (°)	k	α_{95} (°)	polarity	lithology
northern limb												
BZ01	32.08	106.78										
			1	7/8	27.7	34.2	25.8	29.0	14.0	16.7	N	silt-sandstone
			2	0/2	---	---	---	---	---	---		sandstone
			3	3/4*	19.6	19.9	18.9	14.4	1.9	144.4	N	fine-grained sandstone
			4	0/2	---	---	---	---	---	---		sandstone
BZ03	32.07	106.78										
			1	3/4	16.7	29.2	14.0	22.3	150.6	10.1	N	siltstone
			2	3/3	45.1	27.0	41.2	23.6	172.2	9.4	N	siltstone
			3	3/3	14.2	42.4	9.9	35.2	56.8	16.5	N	siltstone
			4	3/3	2.2	39.9	-0.5	31.7	54.5	16.9	N	siltstone
			5	3/3	24.4	43.0	18.9	36.7	212.5	8.5	N	siltstone
BZ05	32.07	106.78										
			1	3/3	199.2	7.4	200.6	10.5	25.1	25.1	R	silt-sandstone
			2	3/3	221.2	-13.6	218.7	-13.9	186.8	9.0	R	siltstone
			3	0/2	---	---	---	---	---	---		sandstone
			4	0/2	---	---	---	---	---	---		sandstone
			5	3/3	208.4	-21.4	204.7	-19.4	69.4	14.9	R	silt-sandstone
			6	3/3	227.1	-49.0	215.4	-49.6	28.0	23.8	R	silt-sandstone
BZ07	32.07	106.77										
			1	3/3	236.2	-55.6	214.6	-49.2	81.4	13.8	R	silt-sandstone
			2	2/2*	173.2	-56.8	168.1	-39.9	---	---	R	silt-sandstone
			3	0/1	---	---	---	---	---	---		sandstone
			4	3/3	44.0	23.0	38.4	15.5	34.8	21.2	N	fine-grained sandstone
			5	4/4	46.6	56.6	26.7	47.8	7.1	37.2	N	silt-sandstone
			6	3/4	27.1	41.6	17.9	29.5	25.1	25.1	N	silt-sandstone
			7	3/3	18.0	35.3	12.0	21.7	255.8	7.7	N	silt-sandstone
BZ09	32.05	106.78										
			1	1/1*	34.5	53.5	25.5	39.9	---	---	N	silt-sandstone
			2	3/3	350.8	40.8	352.0	25.0	243.9	7.9	N	fine-grained sandstone
			3	3/3	331.5	33.3	334.8	18.7	14.3	33.8	N	fine-grained sandstone
			4	4/6	353.3	51.5	354.3	35.6	59.3	12.0	N	silt-sandstone
			5	3/3	23.7	58.5	16.3	43.6	15.8	32.1	N	silt-sandstone
			6	3/3*	342.7	-33.6	338.4	-48.8	1.7	180.0	N	silt-sandstone
BZ11	32.05	106.78										
			1	2/6*	222.1	-20.6	216.2	-15.9	---	---	R	silt-sandstone
			2	1/5*	212.9	-43.9	199.3	-35.1	---	---	R	silt-sandstone
			3	0/1	---	---	---	---	---	---		sandstone
			4	0/2	---	---	---	---	---	---		sandstone
			5	0/2	---	---	---	---	---	---		sandstone
			6	3/3	31.2	48.7	15.6	39.1	26.0	24.7	N	fine-grained sandstone
			7	3/3	1.0	46.7	352.8	31.1	32.4	22.0	N	silt-sandstone
BZ13	32.04	106.78										
			1	0/2	---	---	---	---	---	---		sandstone
			2	0/2	---	---	---	---	---	---		sandstone
			3	0/1	---	---	---	---	---	---		sandstone
			4	4/8*	29.4	-14.3	33.1	-13.5	2.1	88.0	N	fine-grained sandstone
BZ15	32.04	106.78										
			1	0/1	---	---	---	---	---	---		sandstone
			2	3/3	215.2	-42.7	208.6	-39.7	12.2	36.9	R	siltstone
			3	0/2	---	---	---	---	---	---		sandstone
			4	3/3	190.4	-24.6	188.2	-19.1	14.8	33.3	R	siltstone
			5	3/4	201.9	-15.8	200.2	-11.6	28.9	23.4	R	silt-sandstone

The magnetization directions of the high-temperature component are listed. Sampling localities and layers are stratigraphically ordered, in descending order. N and n are the number of specimens in each layer and the number specimens used for the statistics, respectively. Dec. = declination; Inc. = inclination; k = precision parameter; α_{95} = radius of the 95 per cent confidence circle.
mean (1): mean directions are calculated on the basis of all 45 layers.
mean (2): mean directions are calculated on the basis of 36 layers selected from 45. See text for details.
mean (3): the mean direction is calculated on the basis of 11 out of 12 sampling localities. A palaeomagnetic direction of each locality is calculated from ChRM directions of specimens in the locality. One locality (BZ13) is discarded because the circle of 95 per cent confidence is large ($\alpha_{95} = 88.0°$): the other sites have the circle of 95 per cent confidence less than 22.0°.

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Table 1. (*Continued.*)

Locality	Lat. (°N)	Long. (°E)	layer	n/N	In-situ Dec. (°)	In-situ Inc. (°)	Tilt-corrected Dec. (°)	Tilt-corrected Inc. (°)	k	α_{95} (°)	polarity	lithology
southern limb												
BZ23	31.92	106.75										
			1	0/1	---	---	---	---	---	---		sandstone
			2	3/3	194.9	-1.6	197.3	-24.5	40.0	19.7	R	siltstone
			3	0/2	---	---	---	---	---	---		sandstone
			4	0/2	---	---	---	---	---	---		sandstone
			5	0/2	---	---	---	---	---	---		sandstone
			6	4/7	202.0	-3.6	205.4	-25.0	21.9	20.1	R	fine-grained sandstone
BZ21	31.93	106.76										
			1	3/3	181.1	-6.0	181.8	-25.9	27.0	24.2	R	silt-sandstone
			2	0/1	---	---	---	---	---	---		sandstone
			3	3/3	201.8	8.5	201.9	-9.5	102.0	12.3	R	siltstone
			4	6/7	185.3	-1.8	186.0	-21.4	35.4	11.4	R	siltstone
BZ19	31.93	106.75										
			1	0/2	---	---	---	---	---	---		sandstone
			2	4/7	28.7	19.7	33.3	36.1	6.3	40.0	N	fine-grained sandstone
			3	3/3	32.6	31.8	41.2	47.3	105.2	12.1	N	siltstone
			4	3/3	27.0	11.8	30.0	28.6	112.8	11.7	N	siltstone
			5	3/3	18.7	5.4	20.1	23.2	216.6	8.4	N	silt-sandstone
			6	1/1*	33.9	9.4	36.9	24.9	---	---	N	silt-sandstone
BZ17	31.93	106.75										
			1	2/2*	195.5	-7.0	200.2	-22.9	---	---	R	silt-sandstone
			2	0/2	---	---	---	---	---	---		sandstone
			3	0/2	---	---	---	---	---	---		sandstone
			4	0/2	---	---	---	---	---	---		sandstone
			5	3/3	26.4	16.7	35.1	28.4	138.7	10.5	N	siltstone
			6	4/6	40.7	17.4	49.4	23.6	14.3	25.2	N	fine-grained sandstone
			7	4/4	35.2	18.0	44.3	26.2	13.6	25.8	N	fine-grained sandstone
mean (1): all 45 layers												
All	31.8	106.7		45/67	22.3	26.9			9.9	7.1		
							19.9	26.5	13.0	6.1		
Normal				28/67	21.8	30.6			9.9	9.1		
							19.7	27.7	10.8	8.7		
Reversed				17/67	203.1	-20.9			10.0	11.9		
							200.3	-24.8	18.2	8.6		
mean (2): selected 36 layers												
All	31.8	106.7		36/67	22.4	28.6			11.8	7.2		
							20.0	28.8	18.1	5.8		
Normal				23/67	22.1	34.9			15.2	8.0		
							19.3	31.8	18.8	7.2		
Reversed				13/67	202.9	-17.2			10.5	13.4		
							201.3	-23.3	17.9	10.1		
mean (3): 11 samling localities												
All	31.8	106.7		11/12	21.9	26.0			18.7	10.8		
							19.9	27.2	44.1	6.9		

the Yangtze craton is cut by a great sinistral strike-slip fault (Fig. 7a). The Xian-Shui-He fault zone, observed in the SongPan-Ganzi fold belt, extends into the Yangtze craton and separates the southern part of the Yangtze craton from the main part of the Yangtze craton. The Red River dextral strike-slip fault distinguishes it from the Shan-Thai and Simao terranes (Allen *et al.* 1984). The southern part of the Yangtze craton forms a sub-independent block between great sinistral and dextral strike-slip faults.

The large southward movement of the Huili area is interpreted as the extrusion of the whole block of the southern part of the Yangtze craton (Fig. 7b). Avouac & Tapponnier (1993) argue for the possibility of the extrusion of tectonic blocks from Tibet to South China. Realistic models are claimed for the southward extrusion of the southern part of the Yangtze craton with respect to the main part of the Yangtze craton due to left-lateral strike-slip movements along the Xian-Shui-He fault zone and the Red River fault (e.g. Yang & Besse 1993; Roger *et al.* 1995). The southward extrusion of the Huili area followed the large southward movement of Indochina. The southward movement of the Indochina block is suggested by Cretaceous palaeomagnetic studies in the Khorat and Lanping basins in the Indochina block (Yang & Besse 1993; Sato *et al.* 1999).

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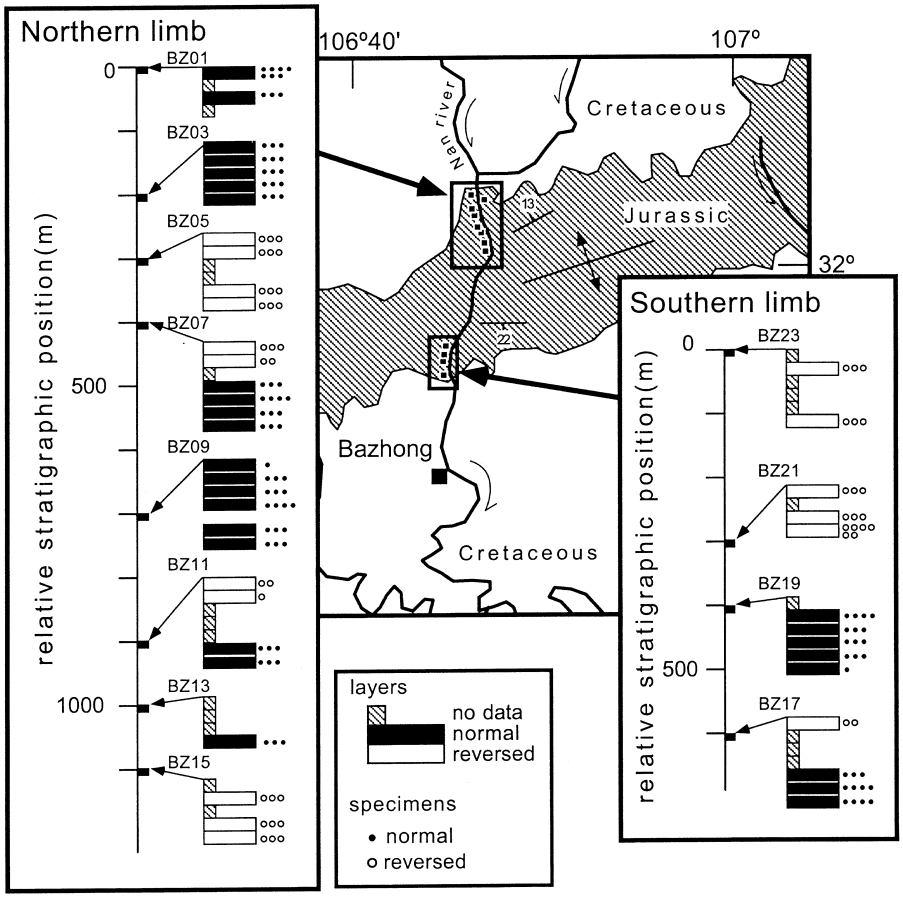


Figure 4. Stratigraphic position of sampling localities (solid squares) and magnetic polarity of the characteristic remanent magnetization for each layer. Specimens with normal (reversed) polarity are shown by closed (open) circles, and the polarity of each layer is denoted by either a black (normal) or white (reversed) column.

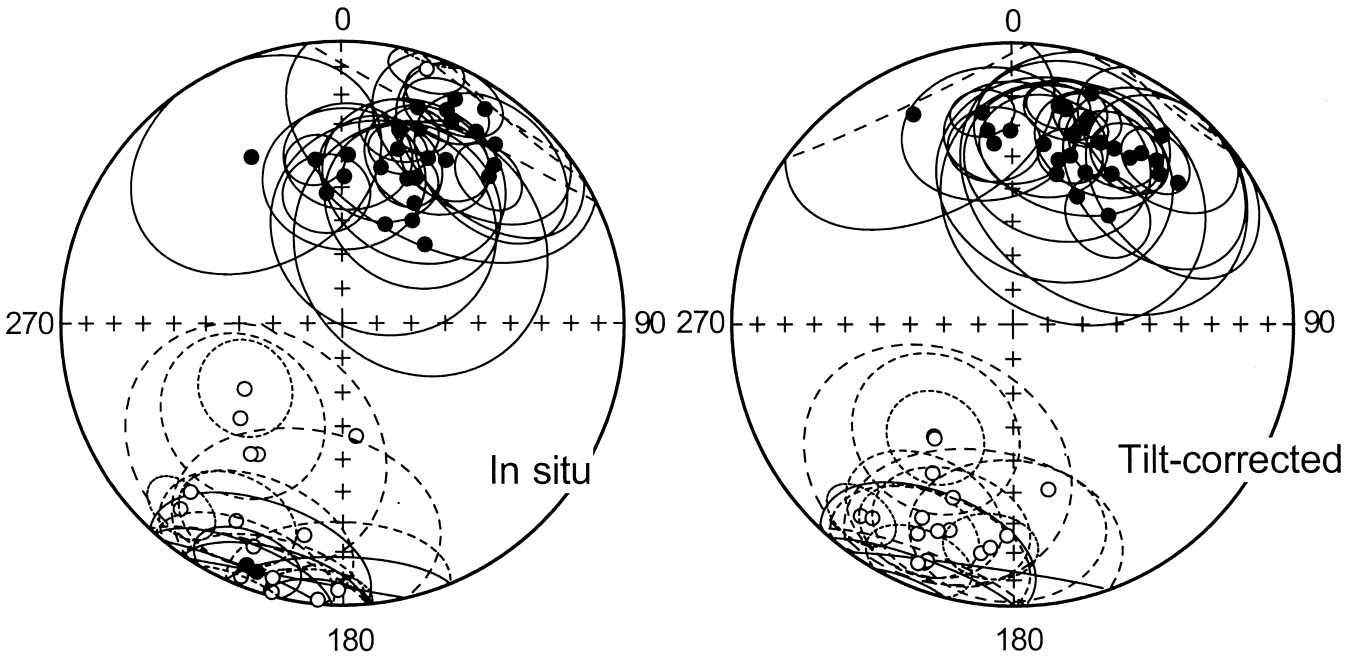


Figure 5. Equal-area projection of the palaeomagnetic directions of selected layers in Table 1 both before and after tilt correction. The mean direction is $D = 24.4°$, $I = 28.6°$ with $\alpha_{95} = 7.2°$, $k = 11.8$ *in situ*, whereas it is $D = 20.0°$, $I = 28.0°$ with $\alpha_{95} = 5.8°$, $k = 18.1$ after tilt correction. The palaeomagnetic direction for each selected layer is drawn as a circle with a circle of 95 per cent confidence; solid (open) symbols on the lower (upper) hemisphere.

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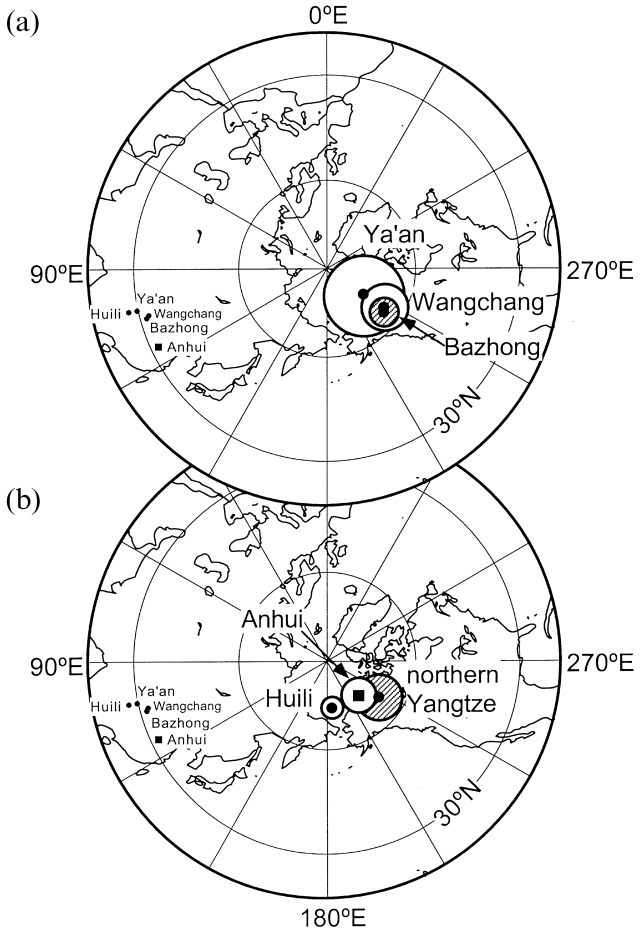


Figure 6. (a) Late Jurassic palaeomagnetic poles from the northern part of the Yangtze craton, with their circles of 95 per cent confidence. These poles are from Bazhong (shaded: this study), Wangchang (Bai *et al.* 1998) and Ya'an (Enkin *et al.* 1991). (b) The mean Late Jurassic pole (shaded) for the northern part of the Yangtze craton calculated from the poles for Bazhong, Ya'an and Wangchang, the pole of the NCB (a solid square) (Gilder & Courtillot 1997) and the pole from the southern part of the Yangtze craton (solid circle) (Huang & Opdyke 1991, 1993).

Subsidiary southward displacement is accomplished by extension of the bounding blocks within the southern part of the Yangtze craton (Fig. 7c). The southern part of the Yangtze craton is cut into several slivers by at least five splays of the left-lateral Xian-Shui-He fault system (Institute of Geology, Chinese Academy of Geological Sciences 1983). Left-lateral slip occurs at the splay faults between slivers associated with left-lateral movement of the Xian-Shui-He fault system. The slip leads to southward extension due to the lateral extrusion of slivers (England & Molnar 1990).

Cretaceous palaeomagnetic data from the northern and the southern parts of the Yangtze craton imply southward displacement of the southern part of the Yangtze craton (Otofuji *et al.* 1998): the observed palaeolatitude for the Huili area is larger than the expected ones calculated from the Ya'an poles, although differences are insignificant ($10.4^\circ \pm 12.4^\circ$, $3.7^\circ \pm 6.6^\circ$). The southward movement of the Huili area probably occurred after the Cretaceous, and was probably caused by the collision of the Indian continent with the Asian continent (e.g. Tapponnier *et al.* 1982).

The Late Jurassic pole for the northern part of the Yangtze craton is compared with the Late Jurassic pole from the NCB (Fig. 6b). A reliable NCB pole is reported for Anhui province (31.6°N, 116.0°E) by Gilder & Courtillot (1997). The pole for the northern part of the Yangtze craton is indistinguishable from the pole of Anhui at the 95 per cent confidence level. This suggests that the amalgamation between the northern part of the Yangtze craton and the NCB ceased before the Late Jurassic, as proposed by Huang & Opdyke (1991) and Gilder & Courtillot (1997). We conclude that the northern part of the Yangtze craton has behaved as a rigid unit with the NCB since the Late Jurassic.

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Table 2. Upper Jurasic palaeomagnetic poles from the North China block and the Yangtze craton.

Locality	Lat. (°N)	Long. (°E)	Pole position Lat. (°N)	Pole position Long. (°E)	A_{95} (°)	N	reference
the North China Block							
Anhui	31.6	116.0	74.4	222.8	5.9	10	Gilder and Courtillot (1997)
northern part of the Yangtze craton							
Bazhong	31.8	106.7	64.7	236.0	7.0	11	this study
Wangchang	32.1	106.1	66.6	236.4	7.6	6	Bai et al. (1998)
Ya'an	29.9	102.9	74.6	235.3	13.6	5	Enkin et al. (1991)
Mean pole			68.6	236.0	8.0	3	
southern part of the Yangtze craton							
Huili	26.8	102.3	74.4	185.1	3.9	20	Huang and Opdyke (1991,1993)

A_{95} = radius of the 95 per cent confidence circle; N = number of sites.

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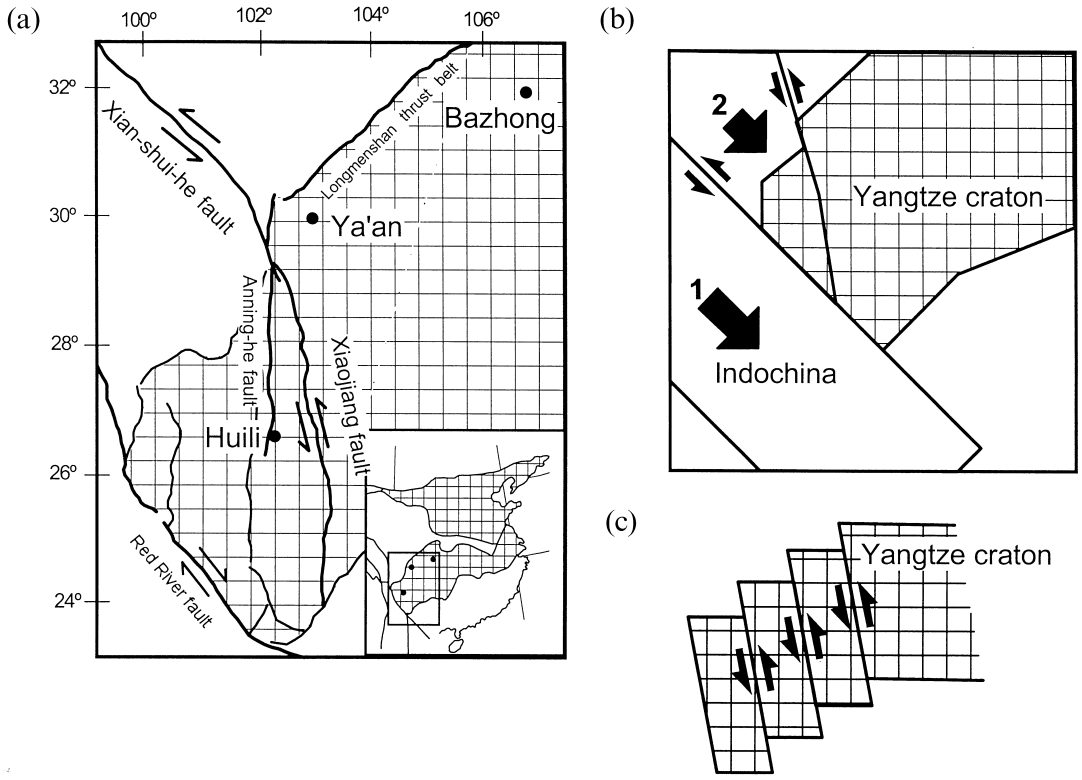


Figure 7. (a) Map of major faults around the southern part of the Yangtze craton. (b) A southward extrusion model of the southern part of the Yangtze craton. This model includes the southward movement of the Indochina block (e.g. Yang & Besse 1993; Roger *et al.* 1995; Sato *et al.* 1999). The large southward movement (1) of Indochina preceded the southward extrusion (2) of the Huili area. (c) A southward extension model within the southern part of the Yangtze craton by lateral movement of slivers. A southward translation of the sliver on the southwest side of each fault occurs relative to the sliver on the northeast side.

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